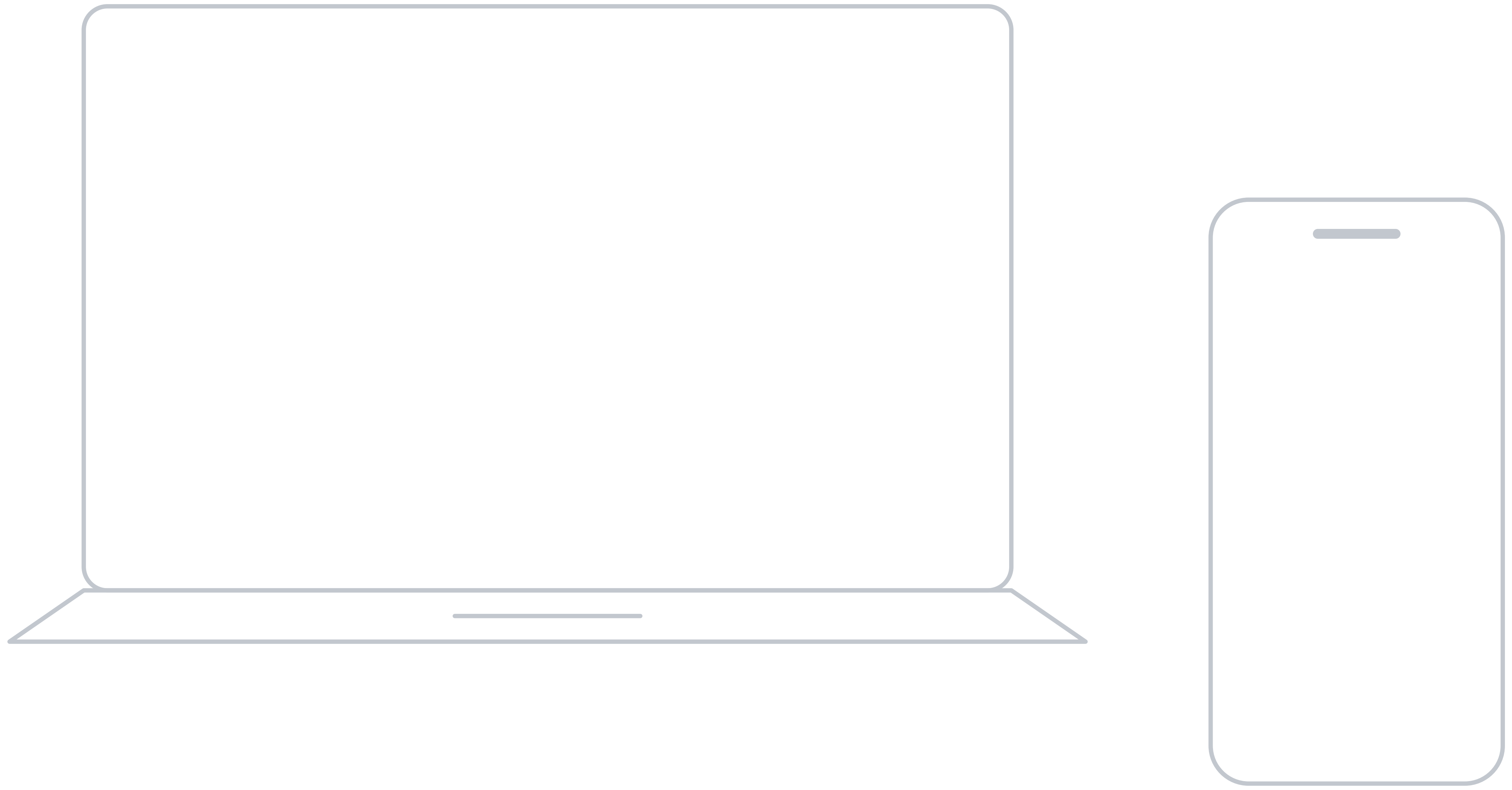
# One model.
# Many behaviors.

Local AI is bespoke AI.



# The same model should be able to

- write code
- read contracts
- handle finance
- translate
- summarize
- teach
- ...

## and let you choose which ones apply.

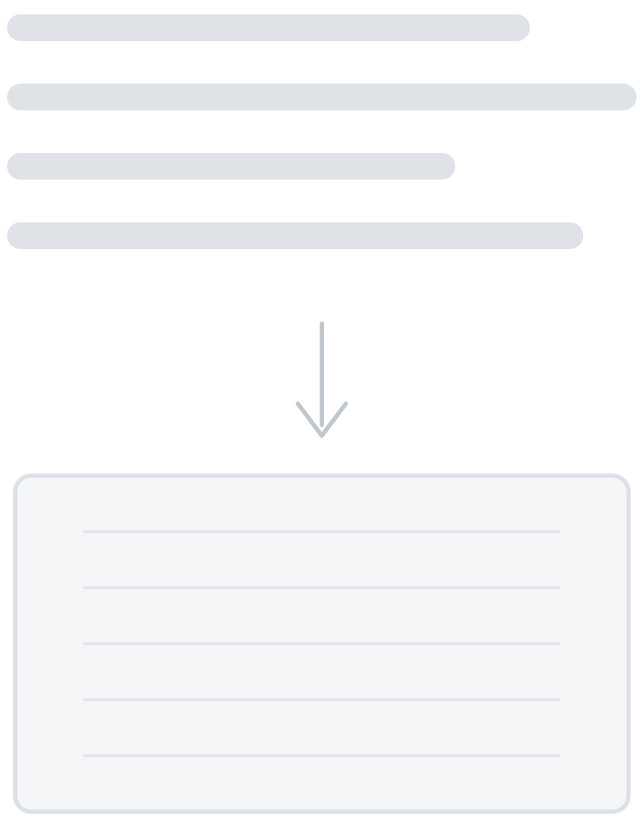
Select the ones you want.
Set how strongly each applies.

# Two things you give a model

## Knowledge

what it knows

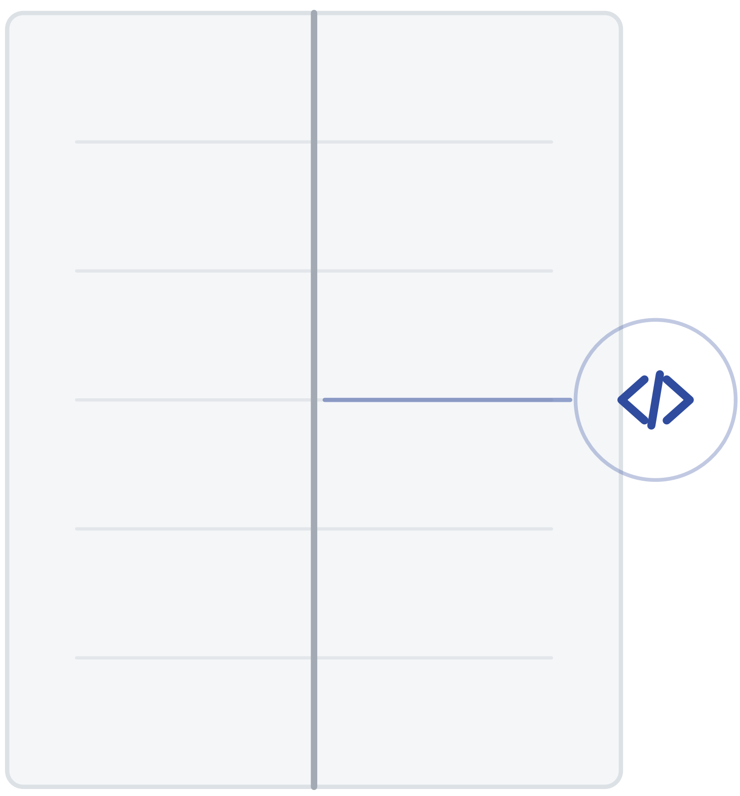
YOU SUPPLY THIS TODAY



## Behavior

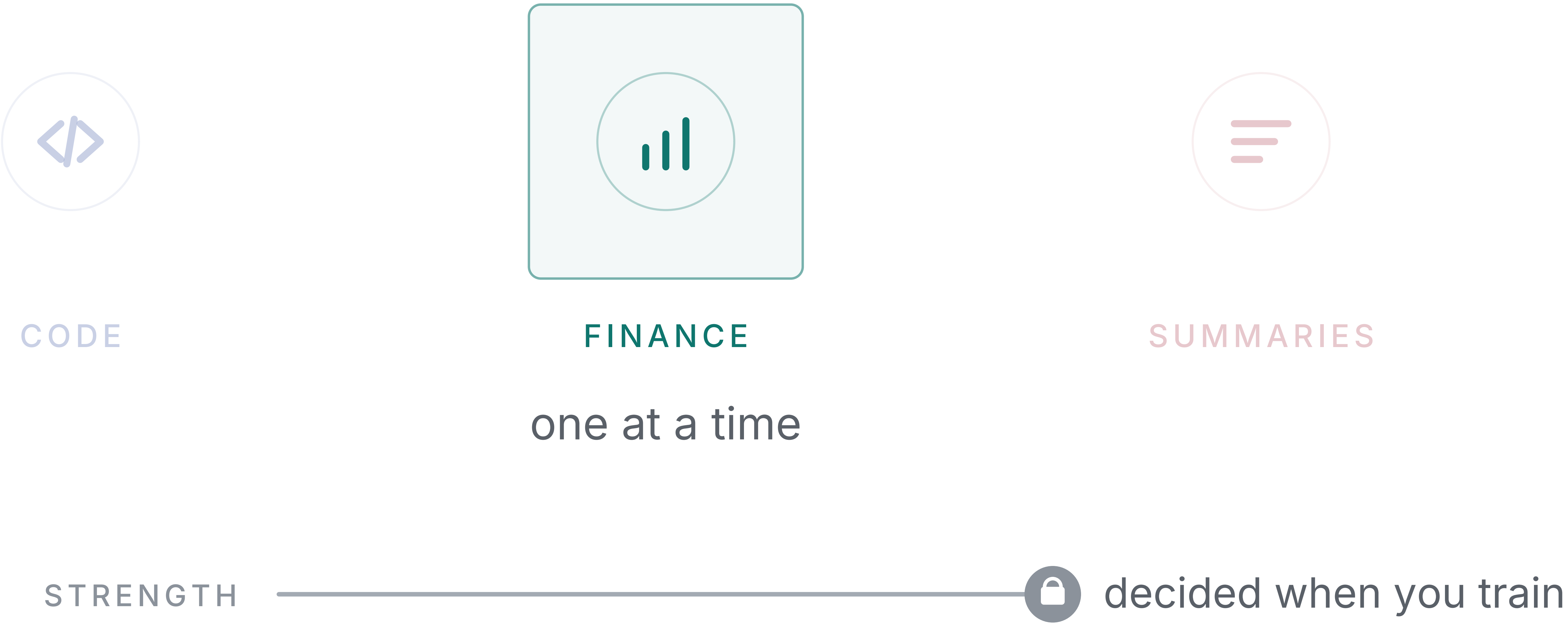
how it responds

YOU CAN SUPPLY THIS TOO



Context made knowledge something you supply.
**Behavior can work the same way.**

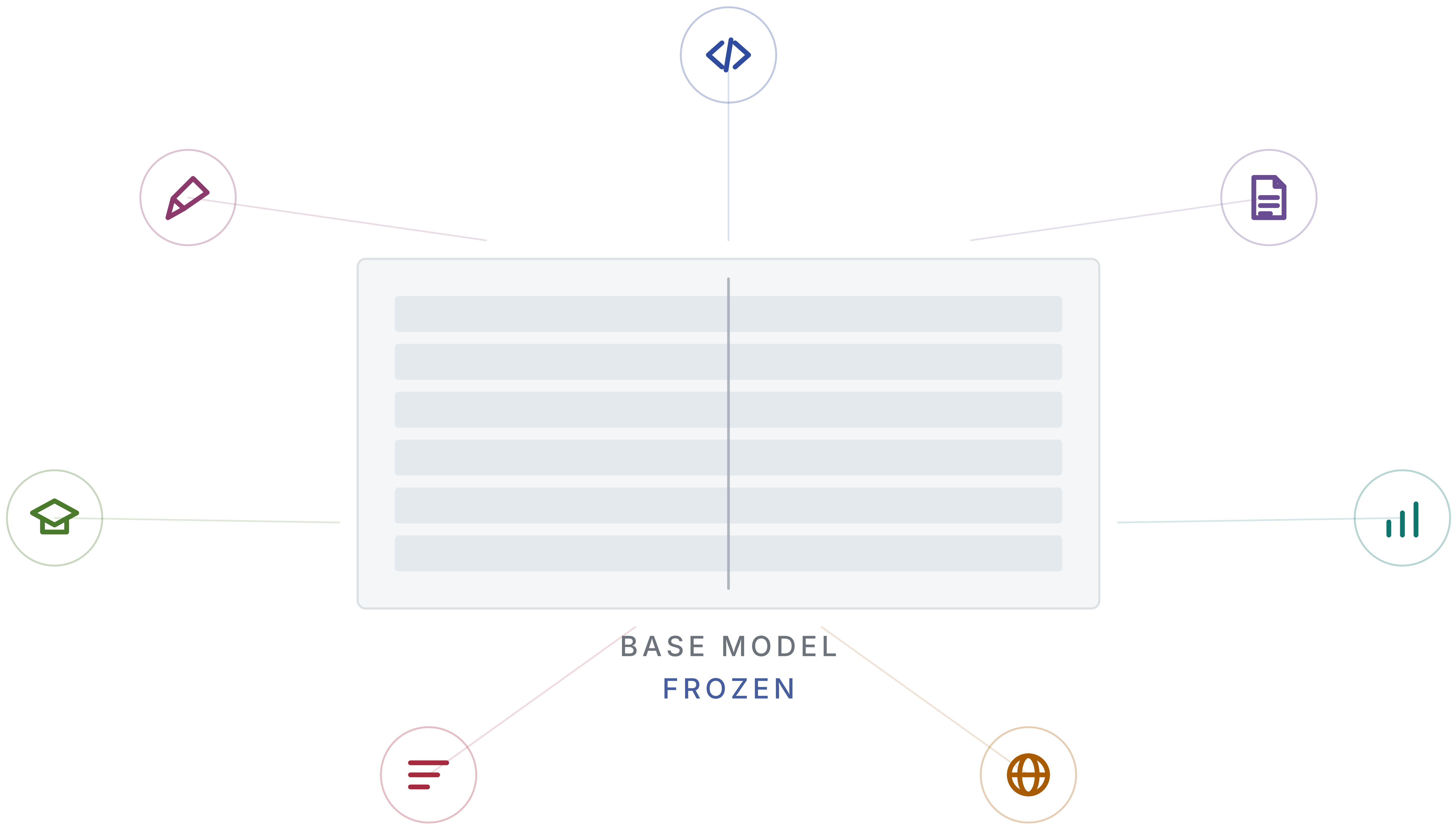
# How adaptation works **today**




A behavior is selected, not continuously composed.
**Its learned effect is fixed when you train.**

# What if it could be?

# A different arrangement




Every behavior stays resident on one frozen model.

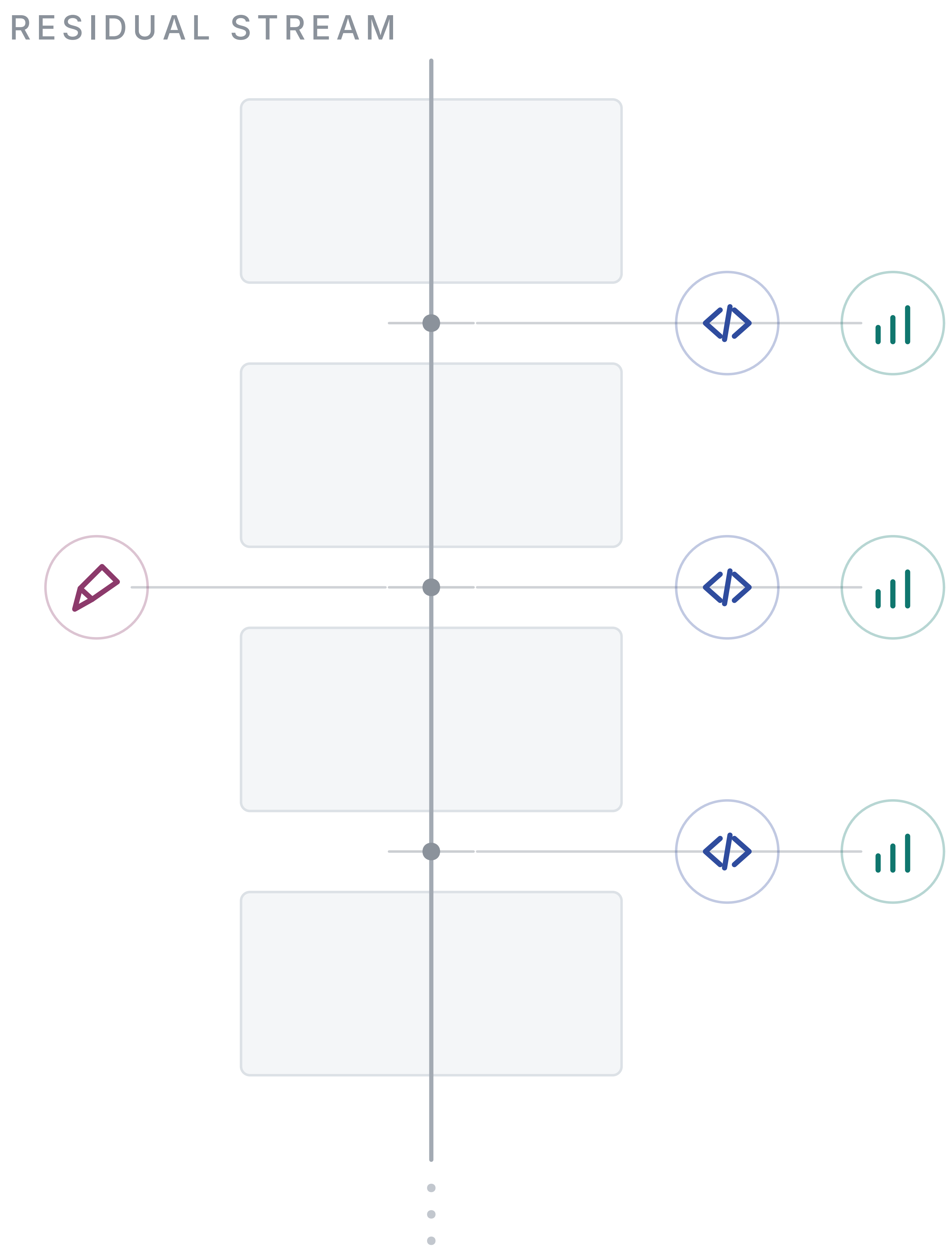
**The base is never written to: no catastrophic forgetting.**

# LARA

## Lightweight Additive Residual Adaptation

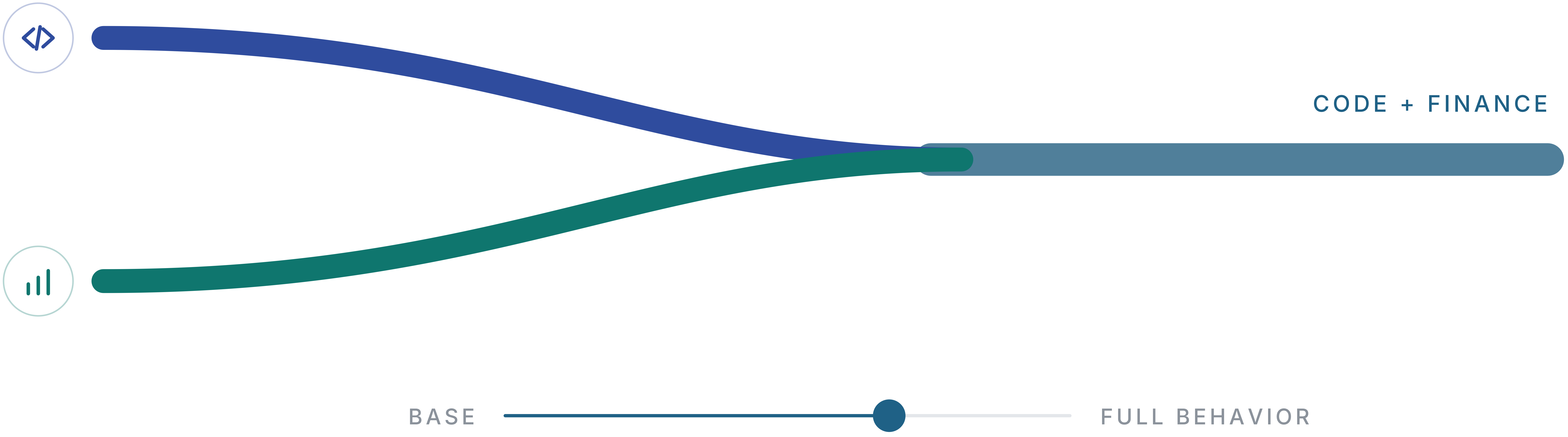
It reads the hidden state between layers,
computes a low-rank correction,
and adds it back to the stream.

**The weights are never written to.**



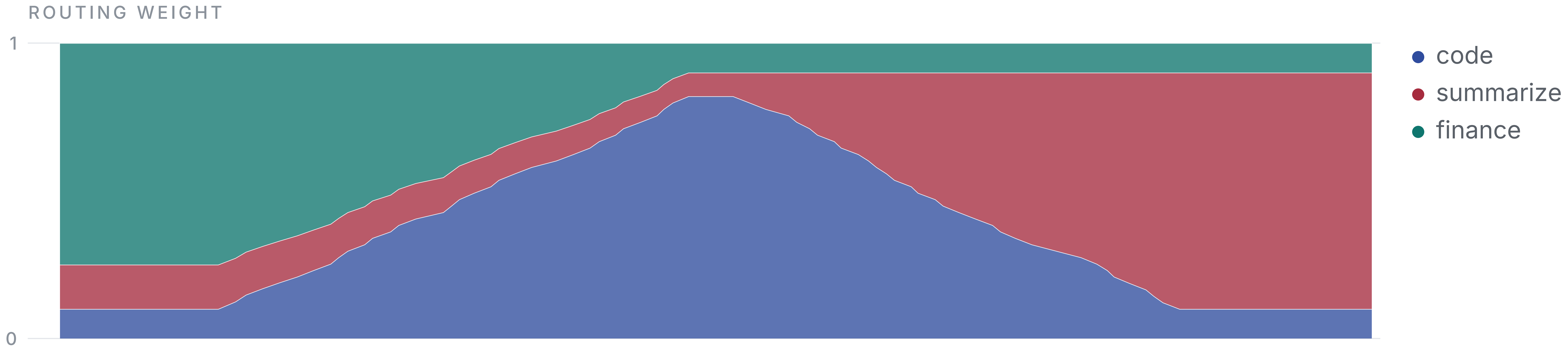

`h ← h + γ · (α/r) · up(down(LayerNorm(h)))`

# Behaviors combine




Strength is a value you set at runtime.
**The same behavior can apply lightly or fully.**

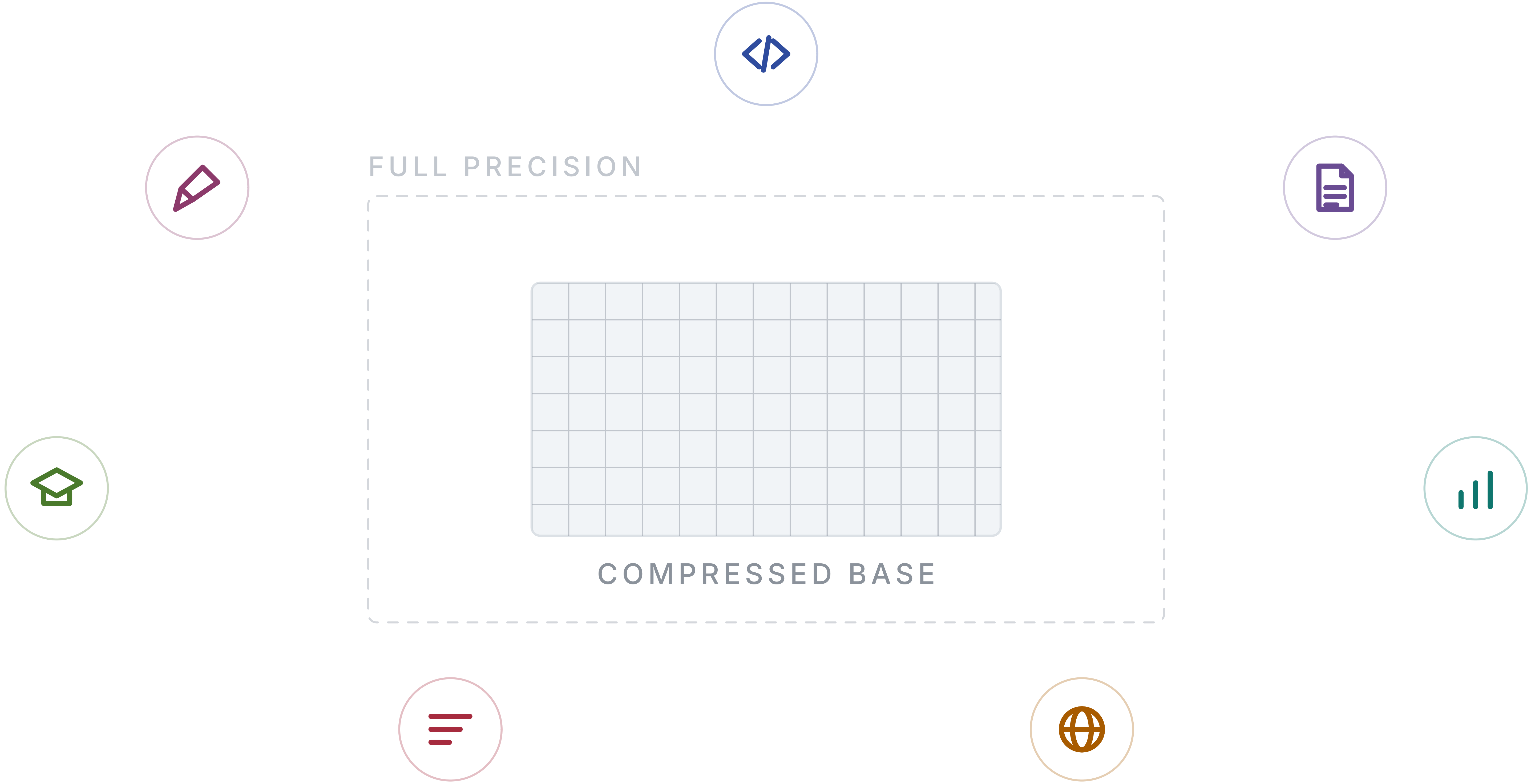
# Routing, per token




The margin dropped to 4.2%, so compute delta = (new - old) / old and summarize the result plainly.

The mix moves as the text is generated.
**No discrete switch between them.**

# It runs on quantized models




The correction never touches the weights,
**so how they are stored does not matter.**

EVERY RESULT IN THE PAPER WAS PRODUCED ON AN 8-BIT BASE

# What each one can do

| | LORA | LARA |
|---|---|---|
| Supervised fine-tuning (SFT) | ✓ | ✓ |
| Preference training (DPO) | ✓ | ✓ 16-28x less memory |
| Reinforcement learning (RL) | ✓ | ✓ 16-28x less memory |
| Behaviors that combine | ✗ | ✓ |
| Strength set when the model runs | ✗ | ✓ |
| Unlimited extensibility | ✗ | ✓ |

Behaviors compose, and they install and uninstall like extensions.
**A behavior is megabytes against a model of gigabytes.**

# One model.
# Many behaviors.

A library you install from.